

Lab Seminar
(Trustworthy AI)

Fairness of Information Flow in Social Networks



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Outline



❖ Introduction

- ❑ Motivation: why information flow fairness?
- ❑ Related work & research gap

❖ Information Unfairness (IUF)

- ❑ Definition, computation and normalization
- ❑ Toy example & comparison with assortativity

❖ MinIUF

- ❑ Reducing unfairness by adding edges

❖ Experiments

- ❑ IUF of real networks, significance tests, MinIUF vs. baselines

❖ Summary & Discussion

Motivation: Why Information Flow Fairness?

❖ Information spreads through social networks

- ❑ People learn about **jobs, promotions, awards** and meet **mentors/sponsors** through their connections
- ❑ Network position shapes who hears what – and who hears it **first**

❖ Structure itself can be unfair

- ❑ E.g., if one group holds the central positions, minorities on the fringe receive less information → inequality reinforces itself

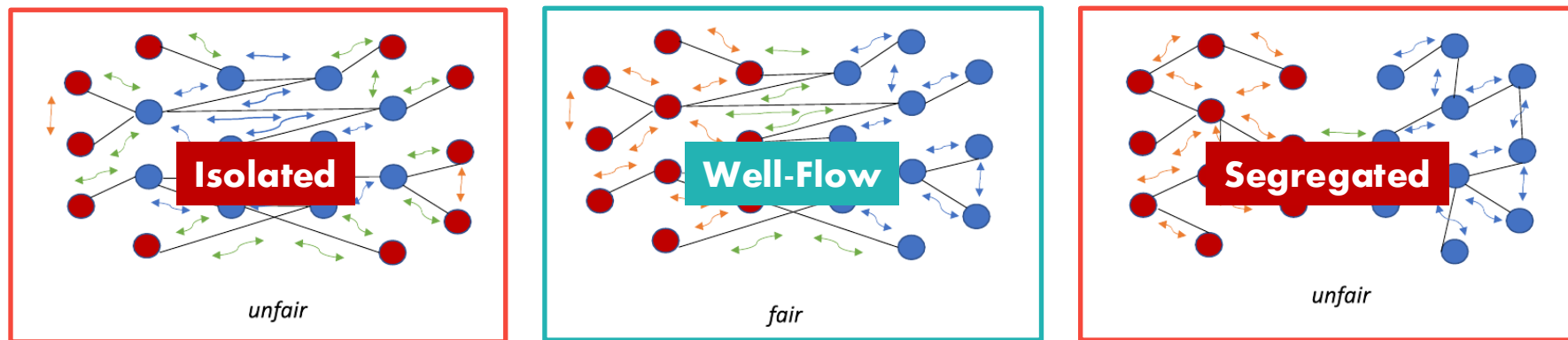


Fig. 1. Same #nodes/#edges per group, different information flow (left: red isolated, middle: fair, right: segregated)

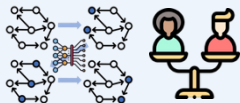
Related Work & Research Gap

Algorithmic Fairness (ML)

Fairness of a model's **outputs**:

credit scoring, sentencing,
node classification, fair clustering

Network is the input,
not the object of study



Fair Influence Maximization

Fish et al., Tsang et al.:

choose **seed nodes** so that groups
receive the spread equally

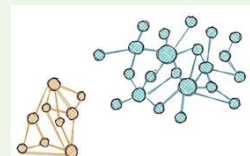
Depends on where information starts



Homophily & Assortativity

Echo chambers, segregation
(Karimi et al., Halberstam & Knight)

Assortativity is a **dyadic** (edge-level)
measure – misses multi-hop flow



This paper: measure whether the **network structure itself** lets protected groups access information equally – regardless of where information starts – and change the structure to reduce unfairness



Fairness of Information Flow in Social Networks

Zeinab S. Jalali, Qilan Chen, Shwetha M. Srikanta, Weixiang Wang, *Myunghwan Kim,
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Syracuse University · *Kumo.AI

ACM Transactions on Knowledge Discovery from Data (TKDD), Vol. 17, No. 6, 2023

Contributions

1

IUF metric

A generalized **Information UnFairness** metric: do different groups have equal access to information spreading in the network?

2

MiniIUF algorithm

Chooses a set of **edges to add** so that IUF of the resulting network is minimized

3

Empirical analysis

IUF of co-authorship, social, e-mail and blog networks; **which structures cause unfairness** and how to reduce it

Extensions over the earlier version (Jalali et al., SDM 2020)

	SDM 2020 (previous)	TKDD 2023 (this paper)
Graph type	Undirected, unweighted only	Any: directed / weighted / simple
Protected groups	Two, non-overlapping	≥ 2 groups, soft (probabilistic) membership
Flow model	Backtracking walks	Non-backtracking walks or SI contagion
Edge-addition algorithm	MaxFair (walks of length exactly k)	MiniIUF (all walk lengths $\leq k$)

IUF: Measuring Unequal Access to Information

❖ Do individuals from different protected groups have different access to information?

- A fair network has $IUF \approx 0$; higher = more unfair. Groups may overlap (soft membership)

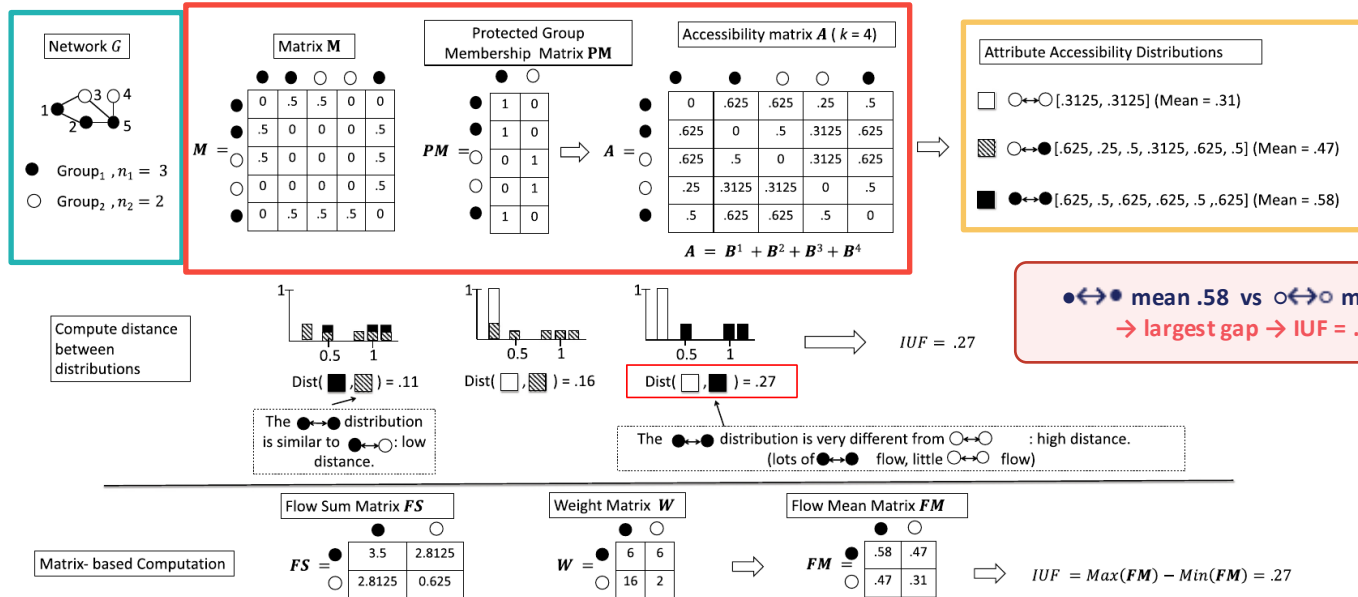


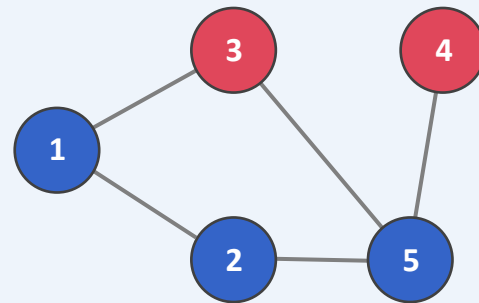
Figure 2: Overview of information unfairness computation process.

Setting and Running Example

❖ Input

- ❑ Graph G with matrix $M : m_{ij}$ = probability that v_i passes information to v_j
 - Directed or undirected, weighted or not
- ❑ Protected groups $P = \{p_1, \dots, p_l\}$, membership matrix PM (can be fractional)
- ❑ K : maximum cascade length — information *expires* (e.g., a job deadline)
- ❑ Optional normalization S ; distance function $Dist$

Running example (used throughout)



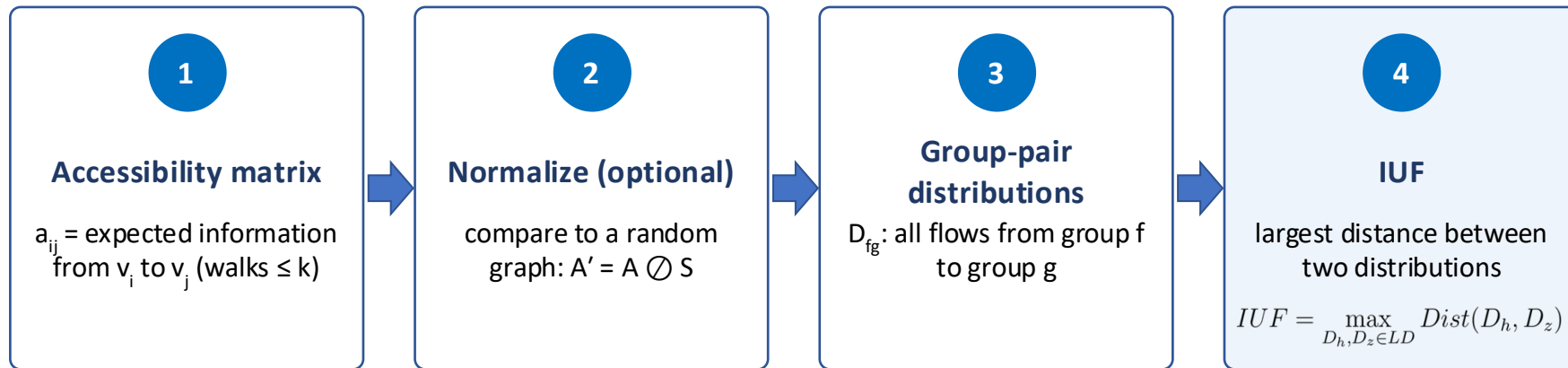
● Group 1 (blue, majority)

● Group 2 (red, protected minority)

every edge: $m_{ij} = 0.5$, $k = 4$

Node 4 (red) has only one link

Information Unfairness (IUF) in Four Steps



MATRIX-BASED COMPUTATION (distance between weighted means, WM)

$$FS = PM^T \cdot A \cdot PM$$

$$W = PM^T \cdot (M - \text{Diag}(M)) \cdot PM$$

$$FM = FS \oslash W$$

$$IUF = \max(FM) - \min(FM)$$

Fair network: $IUF \approx 0$. Higher IUF = some group pair gets much less information

Step 1: How Much Information Flows?

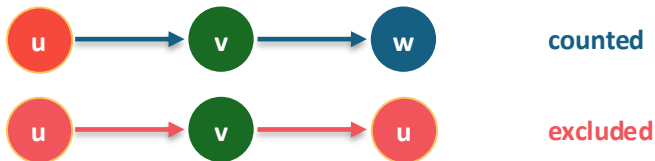
Model 1: Non-backtracking walks

- ☐ a_{ij} = expected times v_j hears information from v_i via walks of length $\leq k$
- ☐ Ignores walks that go straight back ($u \rightarrow v \rightarrow u$)
- ☐ *Fast, deterministic*: $O(kn^3)$



$$\mathbf{A}_{(k)} = \mathbf{B}^1 + \mathbf{B}^2 + \dots + \mathbf{B}^k$$

Why non-backtracking? u hearing back what it told v adds nothing



Model 2: SI contagion (IC model)

- ☐ Each infected node infects a neighbor with probability m_{ij}
- ☐ Monte Carlo $\rightarrow$ probability that information *arrives*
- ☐ More realistic, but *slow* $\rightarrow$ small networks

Walk-based measures the **amount** of information flowing between nodes.

SI-based measures **whether** information is expected to flow

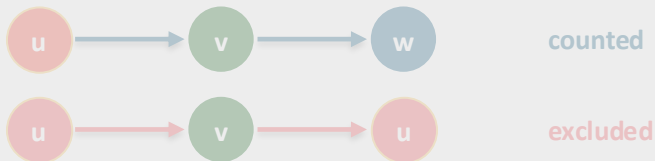
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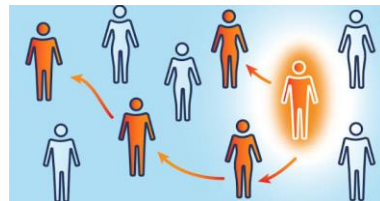
$$A_{(k)} = B^1 + B^2 + \dots + B^k$$

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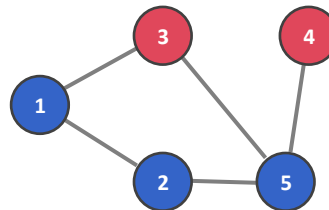
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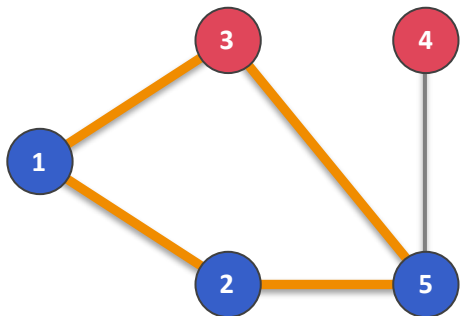
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A (k=4)	1	2	3	4	5
1	0	.625	.625	.25	.5
2	.625	0	.5	.3125	.625
3	.625	.5	0	.3125	.625
4	.25	.3125	.3125	0	.5
5	.5	.625	.625	.5	0

Node 1 $\rightarrow$ node 4: only **0.25** (far away, one link)



Why Do Powers of M Measure Information Flow?



Two 2-hop routes from node 1 to node 5:

$1 \rightarrow 2 \rightarrow 5$ and $1 \rightarrow 3 \rightarrow 5$

One hop: m_{ij} = chance a message crosses edge $i \rightarrow j$ (edges act independently)

Two hops: route $i \rightarrow l \rightarrow j$ delivers with $m_{il} \cdot m_{lj}$; summing over all middle nodes l gives $(M^2)_{ij}$

Example: $(M^2)_{15} = 0.5 \cdot 0.5$ (via 2) + $0.5 \cdot 0.5$ (via 3) = **0.5**

→ node 5 expects 0.5 copies of node 1's message within exactly 2 hops

L hops: $(M^L)_{ij}$ = expected number of copies arriving along length- L walks;
summing $L = 1 \dots k$ gives a_{ij}

a_{ij} counts **expected copies**, so it can exceed 1. It measures the **amount** of information, not the probability that it arrives – that is what the SI model is for.

Why Non-backtracking? Two Walk Models Compared

Running example ($k = 4, m = 0.5$)	Backtracking $M + \dots + M^4$	Non-backtracking $B^1 + \dots + B^4$
Flow 1 $\rightarrow$ 5	1.125	0.5
Mean Blue $\rightarrow$ Blue	1.08	0.58
Mean Red $\leftrightarrow$ Blue	0.81	0.47
Mean Red $\rightarrow$ Red	0.56	0.31
IUF	0.52	0.27

What backtracking adds

Echo walks such as **1 $\rightarrow$ 2 $\rightarrow$ 1 $\rightarrow$ 3 $\rightarrow$ 5**: the message returns to its sender and is counted again as “new”

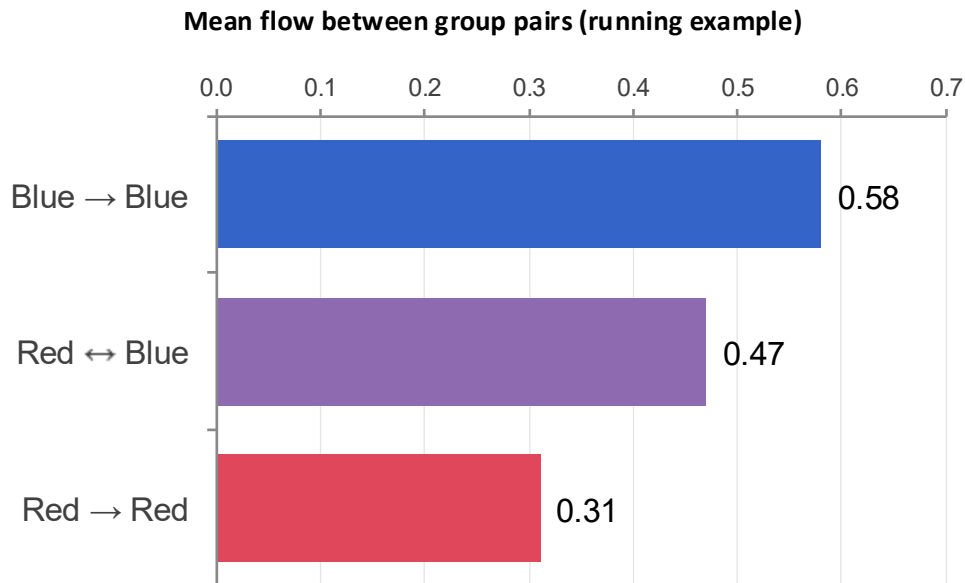
Echoes multiply around well-connected nodes $\rightarrow$ inflate the **advantaged** pairs $\rightarrow$ IUF almost doubles (0.27 $\rightarrow$ 0.52)

How B^k removes them

$B^2 = M \cdot M^T - D$ ($D = \text{diag}(MM^T)$ = the $u \rightarrow v \rightarrow u$ returns)

$B^k = B^{k-1}M + B^{k-2}(I - D)$

Steps 3–4: From Flows to IUF



$$\text{IUF} = 0.58 - 0.31 = 0.27$$

Red gets little information from inside its own group

- Dist = distance between *weighted means* (EMD etc. gave similar results)
- Matrix form (W = number of node pairs):

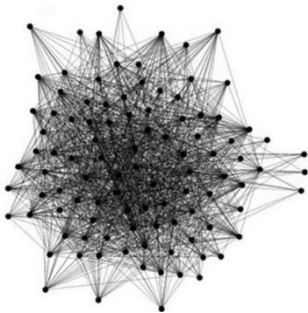
$$\text{FS} = \text{PM}^T \mathbf{A}_{(k)} \text{PM}, \quad \text{FM} = \text{FS} \oslash W$$

$$\text{IUF} = \max(\text{FM}) - \min(\text{FM})$$

Why Do We Need Normalization?

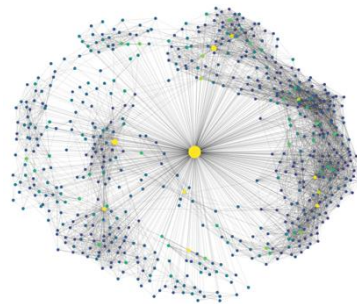
- ❖ A raw flow gap may come from simple structural effects—not only from group placement

A. Density effect



Many edges everywhere → naturally more possible routes

B. Degree effect



Connectivity concentrated on a few nodes → naturally more flow through them

Is the observed flow disparity really due to group structure?

→ Compare observed flow with the flow expected under an appropriate *random-network baseline*.

Two Normalizations, Two Different Questions

❖ Compare With a Random Graph of the Same Kind

- Each flow is divided by the *expected flow* in a random graph with the same property

$$\mathbf{A}'_{(k)} = \mathbf{A}_{(k)} \oslash \mathbf{S}$$

Density normalization

- Denser graphs naturally have more flow
- Lets us *compare networks of different sizes*

$$\text{density: } m_{ij} = \frac{2m}{n^2}$$

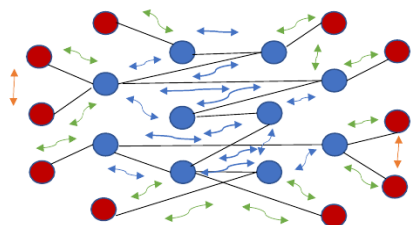
Degree normalization

- Is unfairness *only because one group has fewer links*?
- e.g., senior (high-degree) scientists are mostly men

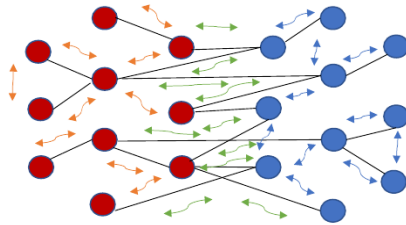
$$\text{degree: } m_{ij} = \frac{d_i d_j}{2m}$$

Paper: degree differences can explain the gap, but this “would not necessarily *excuse* unfairness”

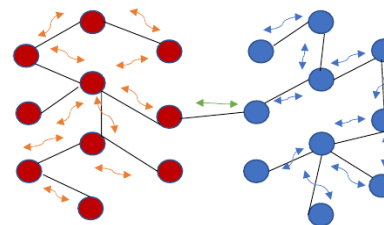
Toy Example: Three Networks of Fig. 1



unfair



fair



unfair

Walk-based ($k = 4, pp = 0.5$)	Left (red isolated)	Middle	Right (segregated)
Mean red-red flow	0.04	0.12	0.17
Mean blue-blue flow	0.18	0.11	0.20
Mean red-blue flow	0.12	0.12	0.08
IUF (walk-based)	0.14	0.01	0.12
IUF (SI-based)	0.18	0.02	0.15
Problem → remedy	red isolated → <i>peer links</i> among red	fair	segregated → <i>links between groups</i>

Both unfair networks get high IUF, but for **different reasons** → different remedies (connect red nodes vs. integrate groups)

What IUF Can See That Homophily Cannot

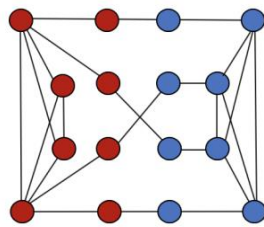
❖ Assortativity (homophily) is related, but **dyadic**

- ❑ Only looks at direct edges, not multi-hop information flow

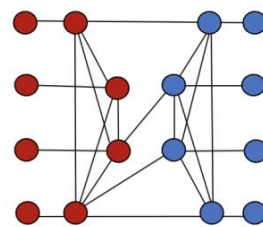
❖ Fig. 3: three graphs with the **same** assortativity (0.67)

- ❑ Information flows easily within red and within blue in all three
- ❑ But in Graph (b) flow from blue to red is much easier
- ❑ **Assortativity cannot see this – IUF can**

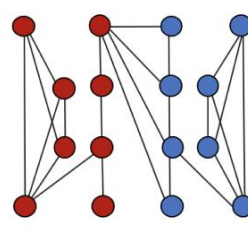
	Graph (a)	Graph (b)	Graph (c)
Assortativity	0.67	0.67	0.67
IUF (walk)	0.64	0.15	0.57
IUF (SI)	0.40	0.12	0.35



(a) Graph(a)



(b) Graph(b)



(c) Graph(c)

Fig. 3. Graphs with same assortativity coefficient values, but different information unfairness scores

MinIUF: Reduce Unfairness by Adding Edges

❖ Problem

- Given G , k , PM and a budget b : add b *new edges* that minimize IUF
- Adding is practical (mailing list, meeting, friend recommendation); removing is not

❖ Challenge: IUF Is Not Submodular

- One edge can *increase* IUF, while several edges together decrease it

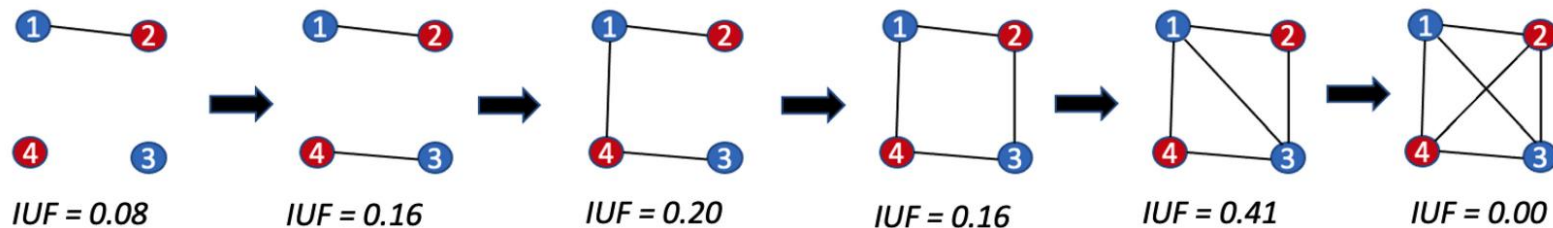
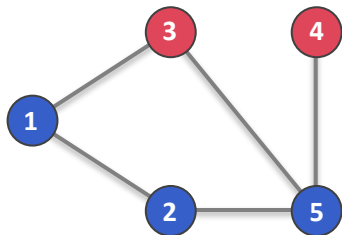


Figure 4: Adding single edge increases IUF while adding multiple edges decreases IUF.

Why Can Adding One Edge Make IUF Worse?

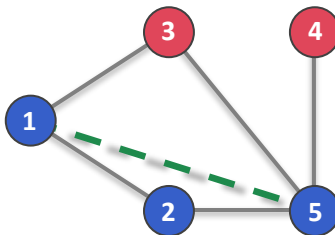
Original



	Blue	Red
Blue	0.58	0.47
Red	0.47	0.31

IUF = 0.27

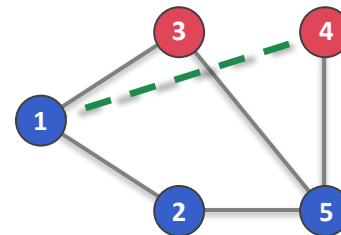
+ edge 1–5 (blue–blue)



	Blue	Red
Blue	1.08	0.76
Red	0.76	0.44

IUF = 0.65 (+138%)

+ edge 1–4 (red–blue)



	Blue	Red
Blue	0.75	0.71
Red	0.71	0.63

IUF = 0.13 (–54%)

IUF = $\max(\text{FM}) - \min(\text{FM})$. Edge 1–5 boosts the **already-largest** entry (0.58→1.08) far more than the smallest (0.31→0.44) → gap widens. Edge 1–4 lifts the **weakest** pairs → gap shrinks. An edge's effect depends on the rest of the graph → **no diminishing returns (not submodular)**

MinIUF: How It Scores a Candidate Edge

- 1 Compute flow matrix FM (all group pairs)
- 2 For every non-edge (u_i, u_j) : estimate new FM' from extra walks of *every length* $< k$ through the new edge
- 3 Add the b / x edges with the lowest score
- 4 Repeat x times

$$score(u_i, u_j) = IUF_{after} - IUF_{before}$$

Algorithm MinIUF

Input: M, k, PM, budget b, iterations x

- 1 counter $\leftarrow 0$
- 2 $A(k), B^k$ (non-backtracking); $AC^k = A(k) \cdot PM, BC^k = B^k \cdot PM$
- 3 FM from $A(k)$ and PM
- 4 $IUF_{before} = \max(FM) - \min(FM)$
- 5 for each pair (v_i, v_j) not in G:
 - $FM'_{fg} = FM_{fg} + pp_{ij} \cdot \Sigma_k'$ (flows of length $\leq k$ through (v_i, v_j))
 - score = $IUF_{after} - IUF_{before}$
- 6 add the b/x best-scoring edges to G
- 7 counter += 1; if counter $< x$ go to 2

Complexity: $O(x(l^2n^2 + kn^3))$ — matrix operations, can use GPUs

Why the Score Formula Looks Like That



Every **new** walk through $(u, v) = \text{walk into } u + \text{the new edge} + \text{walk out of } v$ (total length $\leq k$, both directions $u \leftrightarrow v$)

Why it is cheap

AC = A-PM and BC = B-PM are computed **once**: flow between every node and every group

Each candidate then costs only $O(l^2k)$ – no need to recompute A ($O(kn^3)$) per edge

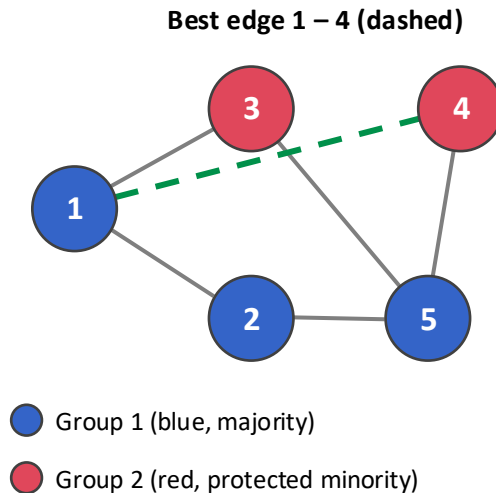
Why it is an estimate

Ignores walks that use the new edge **twice** and interactions between edges added in the same batch

Exact choice when adding one edge per step; batches of b/x trade accuracy for speed

MinIUF on the Running Example

Add edge	Type	IUF after	Change
<i>none</i>	—	0.27	—
1 – 4	red–blue	0.13	-54%
3 – 4	red–red	0.17	-38%
2 – 4	red–blue	0.23	-15%
2 – 3	red–blue	0.52	+92%
1 – 5	blue–blue	0.65	+138%



Connecting the *isolated red node* halves IUF; linking two *blue* nodes makes it worse

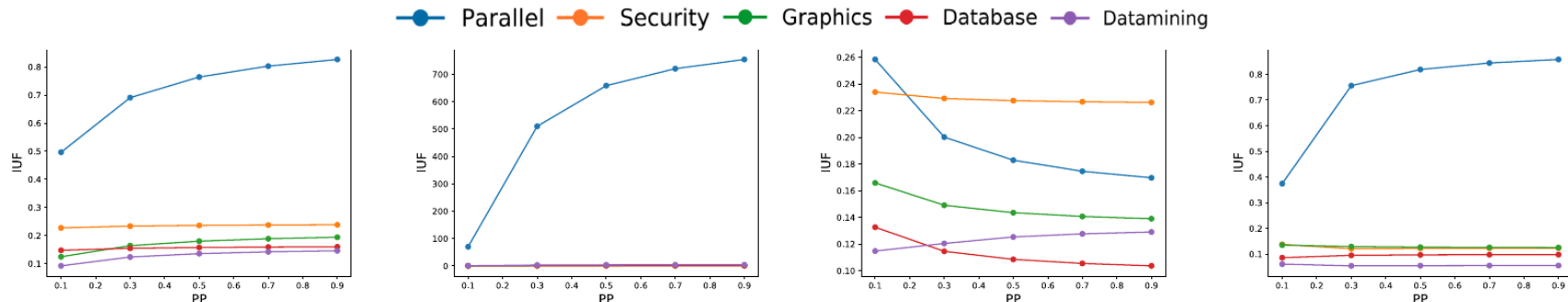
Experimental Setup

Dataset	Type	#Nodes	#Edges	Assort.	Attribute: Group 1 / Group 2
Parallel	Co-authorship (DBLP)	1,251	4,356	0.06	Men 82% / Women 18%
Graphics	Co-authorship (DBLP)	3,525	10,399	0.06	Men 71% / Women 29%
Security	Co-authorship (DBLP)	1,962	5,976	0.07	Men 80% / Women 20%
Databases	Co-authorship (DBLP)	3,185	9,386	0.10	Men 68% / Women 32%
Data Mining	Co-authorship (DBLP)	2,272	7,643	0.08	Men 66% / Women 34%
Pokec-pl / pg	Social network (Slovakia)	3,312	22,707	0.07 / 0.09	Language 84/16% · Gender 50/50%
Pokec-zl / zg	Social network (Slovakia)	3,018	23,470	0.03 / 0.03	Language 85/15% · Gender 51/49%
Enron	E-mail network	144	1,344	0.03	Men 76% / Women 24%
Polblog	Blog directories	1,224	16,715	0.81	Liberal 48% / Conservative 52%

DBLP: top-tier venues 2015–2019, largest connected component; gender inferred with [Gender API](#) (>90% accuracy for most names)

Settings: $k \in \{2, 4, 6, 10\}$, propagation probability $pp \in \{0.1, 0.3, 0.5, 0.9\}$; $m_{ij} = pp$ for every edge

Result 1: Every Network Is Unfair

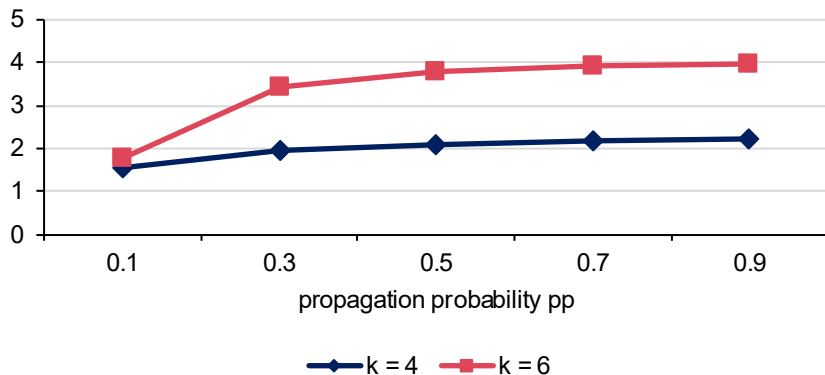


(e) Density-normalized NBT-walk-based method (k=2). (f) Density-normalized NBT-walk-based method (k=6). (g) Degree-normalized NBT-walk-based method (k=2). (h) Degree-normalized NBT-walk-based method (k=6).

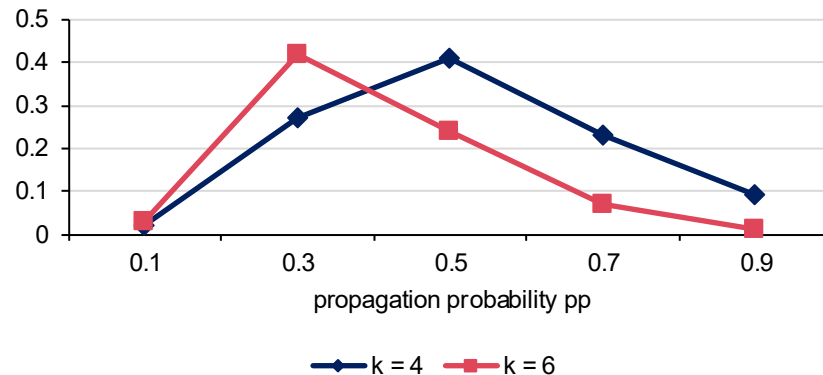
- ❑ **All networks** have IUF > 0 — but for *different reasons*
- ❑ **Parallel**: one clique of 38 men + 1 woman; removing it: IUF 1,409 → 33
- ❑ **Degree-normalized** IUF is much lower → *degree differences explain a large part*
 - Walk-based IUF grows with pp (amount of flow); SI-based IUF shrinks (whether info arrives)

Why Walk-IUF Rises but SI-IUF Falls with pp

Walk-based IUF (density-normalized)



SI-based IUF



Walks: counts keep growing

Expected copies $\propto pp^l \times \text{\#walks}$;
 walks **explode** in dense same-group neighborhoods $\rightarrow$ advantaged pairs grow fastest

SI: probabilities saturate at 1

High pp $\rightarrow$ almost everyone receives everything $\rightarrow$ flows **equalize**;
 longer k saturates sooner

Why Is Parallel So Unfair? Walks Explode in Cliques

Walks of length L between two members of a clique of c nodes

Walk length L	Clique $c = 5$	Clique $c = 39$
2	3	37
4	51	53,465
6	819	77,203,497

Formula: $((c - 1)^L - (-1)^L) / c$ – grows like $(c - 1)^L$

Parallel network

One clique of **38 men + 1 woman**

Its walks dominate Men–Men flow $\rightarrow$ max(FM) explodes

IUF **1,409** $\rightarrow$ **33** after removing that clique ($k = 6$, $pp = 0.4$)

What this tells us

- ✓ Real effect: a tight clique really does circulate information among its members (e.g., large author teams)
- ✓ Degree normalization absorbs much of it – clique members simply have very high degree
- ✗ Also a limitation: a single dense substructure can dominate the whole IUF value

Three Null Models: What Each One Rules Out

Random graphs keep...	Question it answers	Result (Table 3)
Same n and m (Erdős–Rényi)	Is IUF just a matter of size / density?	0.0% exceed → No
Same degree sequence (config. model)	Is IUF just degree differences between groups?	≤ 0.2% exceed → No
Same topology, shuffled group labels	Does it matter where the groups sit?	0.0–53.6% exceed → often not!

How to read a p-value here

p = fraction of the 1,000 random graphs whose IUF $\geq$ the real IUF

$p < 0.05 \rightarrow$ the real network is more unfair than that null model explains

Why the shuffle result is surprising

Random labels on the **same topology** are often just as unfair
 $\rightarrow$ cliques, hubs and skewed degrees make unfairness likely
whoever is in the minority

Is It Significant? Who Is Disadvantaged?

Shuffle test	Density	Degree	Attributes
DBLP (5 subfields)	0.0%	$\leq 0.2\%$	11–54%
Pokec (4)	0.0%	0.0%	0.1–3.8%
Enron	0.0%	0.0%	36.3%

% of 1,000 random graphs with higher IUF ($k = 4$, $pp = 0.5$)

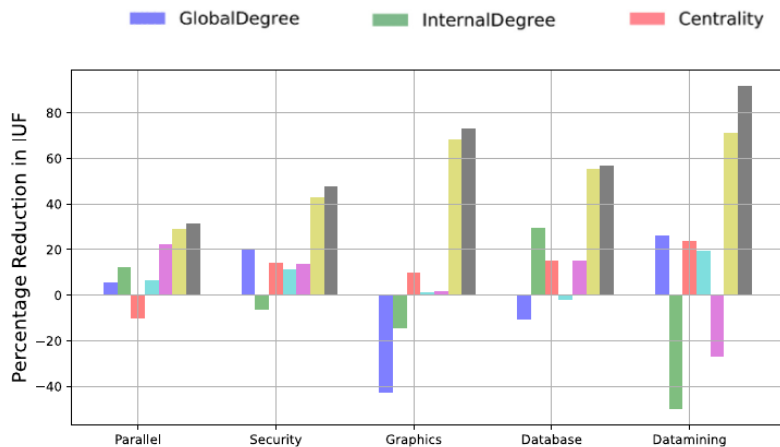
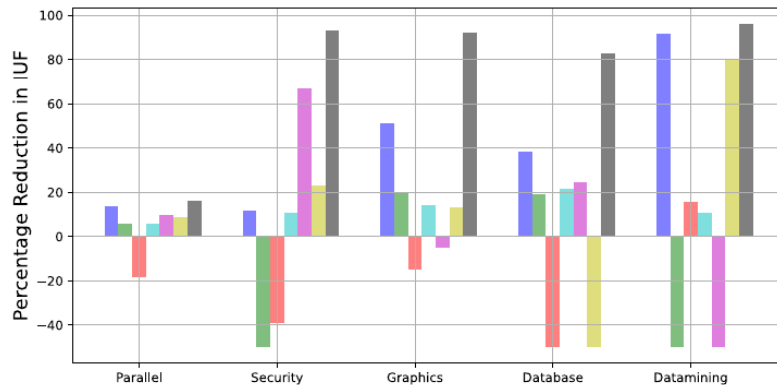
Mean flow	M $\rightarrow$ M	M $\leftrightarrow$ W	W $\rightarrow$ W
Parallel	26.6	6.63	4.05
Data Mining	4.39	4.71	5.04
Pokec-pg	8.98	7.19	6.17
Enron	2.28	2.01	1.77

- ❑ Not explained by density or degree alone
- ❑ Shuffled labels often as unfair $\rightarrow$ *topology itself* (cliques, hubs)

- ❑ M = men, W = women
- ❑ Most co-authorship nets: *W $\rightarrow$ W is highest*
- ❑ Parallel, Pokec, Enron: *women get less*

Unfairness is real, but its *cause differs* by network

Result 2: A Few Edges Remove Most Unfairness

(c) $k=2$, $pp=0.5$.(d) $k=6$, $pp=0.5$.

- Only **1% new edges**: IUF -40% ($k=2$) and **-80–90%** ($k=6$) in most cases
- MinIUF (gray) beats all 6 baselines; some baselines even **increase** IUF

Why Do Some Baselines Make Unfairness Worse?

Centrality / EigenCentrality

Connect the most central nodes – usually in the dense **majority core** → boosts the largest FM entry
→ often **negative** bars

GlobalDegree / InternalDegree

Target the worst group pair, but extra flow through **hubs** spills into every pair → unstable (+/-)

Random

Unfocused edges → small effect

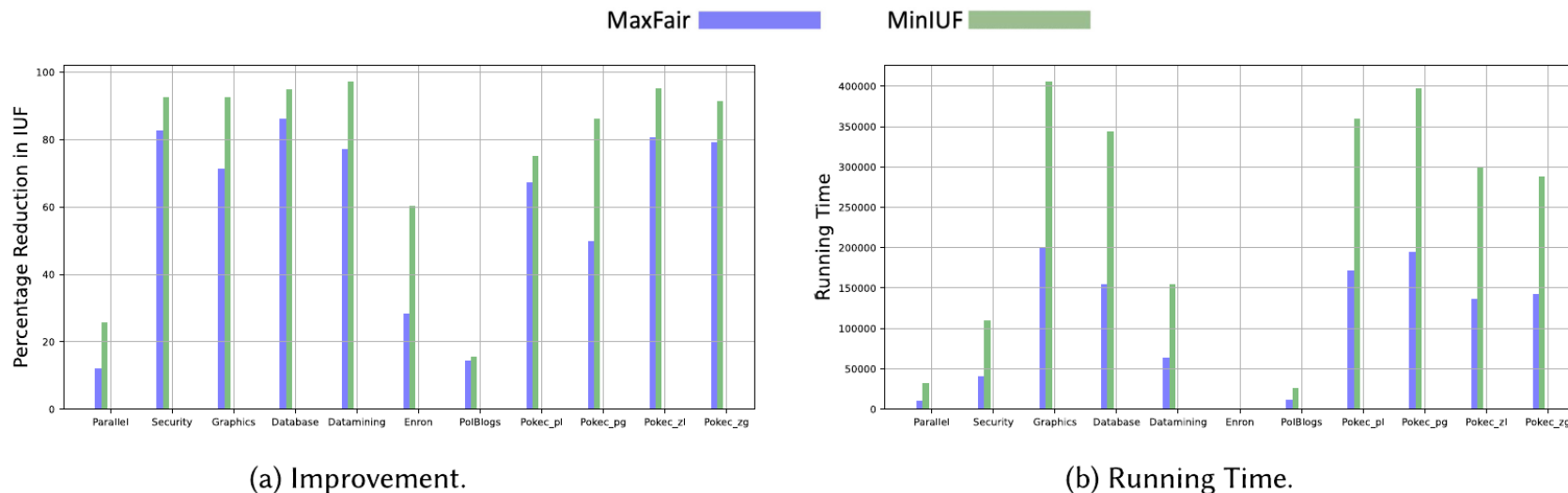
MaxFair

Right target, but only walks of **one length k** → misses shorter walks

MiniUF

Directly estimates **Δ IUF** using walks of **all lengths $\leq k$** → consistently best

Accuracy vs. Running Time



- ❑ **MiniIUF** reduces IUF more; **MaxFair** is faster (same big-O, fewer operations)
- ❑ $x \geq 8$ re-scoring rounds is enough

Summary at a Glance

	Information Flow (TKDD'23)
Question	Equal access to information?
Measure	IUF: gap between group-pair flows
Flow model	Non-backtracking walks or SI cascades
What is fixed	The <i>network</i> (add edges)
Key insight	Fairness depends on how information propagates

Summary

❖ Problem

- ❑ Does the network structure give protected groups equal access to information?

❖ IUF – Information Unfairness metric

- ❑ Accessibility matrix (non-backtracking walks or SI) $\rightarrow$ group-pair flow distributions $\rightarrow$ max distance
- ❑ Density/degree normalization; captures multi-hop effects that assortativity misses

❖ MiniIUF – reducing unfairness by adding edges

- ❑ Scores candidate edges by their effect on flows of all lengths $\leq k$; outperforms 6 baselines

❖ Findings

- ❑ All studied networks are unfair to some degree; degree differences and dense cliques are major causes
- ❑ Adding $\sim 1\%$ edges can reduce IUF by 40% ($k = 2$) to 80–90% ($k = 6$)

Thank You!



"Output is conditional on INPUT."

Normalization: Where Do the Null Models Come From?

Density: Erdős–Rényi random graph

Keep n and m ; every pair is **equally likely** to be connected

m edges spread over $\sim n^2/2$ pairs $\rightarrow P(i-j) \approx 2m / n^2$ for every pair

Running example: $2 \cdot 5 / 5^2 = 0.4$ for all pairs

Degree: configuration model

Node i has d_i “stubs”; the $2m$ stubs are paired **at random**

Chance that i 's stubs meet j 's $\rightarrow P(i-j) \approx d_i d_j / 2m$
(same term as modularity)

Running example: $d_5 = 3, d_4 = 1 \rightarrow 3 \cdot 1 / 10 = 0.3$

Why an estimate instead of sampling?

Averaging accessibility over t random graphs costs $t \times O(kn^3)$. The paper builds S once from the **expected** adjacency matrix; Table 6 shows it is close to the average over 1,000 sampled graphs.